Attention Group Members: This is a group assignment -- the purpose is to collaborate and learn from each other. We expect the group to solve problems together so that everyone benefits from the discussion. It works well if everyone participates. If your group is not working out or you need help, please contact the teaching team.

No late assignments accepted.

Locate your group members in this spreadsheet

Links to an external site.

Group Member names (must be PRESENT)	Where is the meeting taking place?
James Anderson	SNE Room 116
Kate Pun	SNE Room 116
Edwin Huang	SNE Room 116
Yash Rao	SNE Room 116
Younwoo Park	SNE Room 116

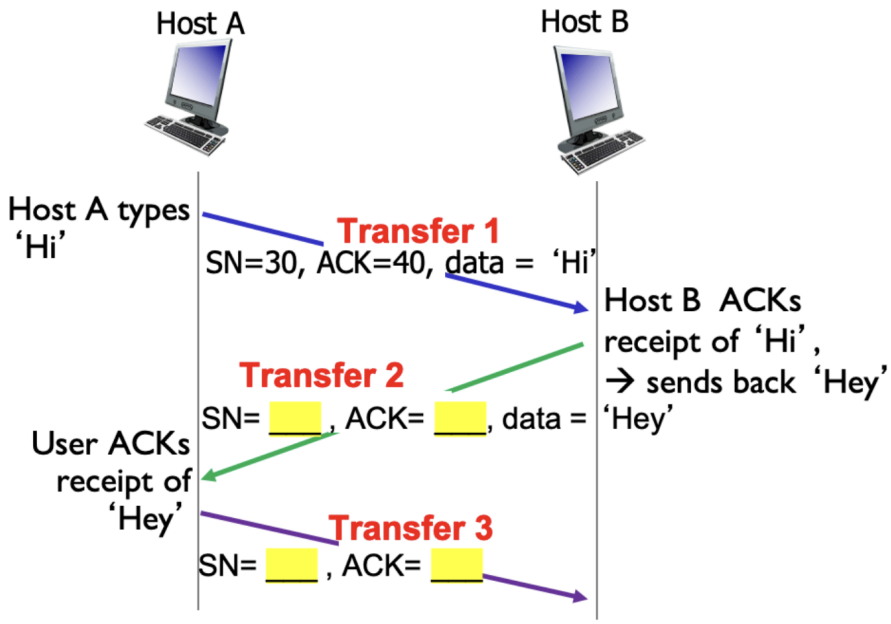
What is the link to your Google doc?

CSE 150 Group Asgn 5

1. TCP Data Transfer: SNs and ACKs

Given the TCP Transfer below, fill in the missing SN and ACK numbers. Assume the following:

- Transfer 1 is 2 bytes of data
- Transfer 2 is 3 bytes of data
- Transfer 3 is 3 bytes of data



Submit the following in the box below:

a) Transfer 2: SN= 40, ACK= 32

b) Transfer 3: SN= 32, ACK = 43

2. TCP Deployment and Congestion Control: Group Discussion - 2 videos

1. Thinking over this video in which Vint Cerf deploys the TCP protocol:

- How did Vint Cerf deploy TCP and what challenges did he face?
    - The challenge that led to the development of TCP was the need to standardize the different packet sizes and speeds being used at the time. Some specific problems that TCP had to handle were duplicate detection and out of order delivery of packets. In order to deploy TCP, he announced that by January 1, 1983, arpanet would be switching from NCP to TCP/IP. However, he faced some challenges enforcing this change because it required coordination with multiple parties. To help incentivize the switch, he disallowed NCP packets and only allowed TCP/IP on ARPANET for a day.

- Would it be possible to deploy a new version of TCP today? What would be the challenges?
    - It would be possible to deploy a new version of TCP, but there would be many challenges. The new TCP protocol must be backward compatible to

accommodate the billions of devices and numerous networks running existing TCP stacks. The modern internet is also massive and updating or replacing the protocol would be extremely challenging and maybe even impossible. There would be issues with cost, security risks, and service disruptions.

Discuss and explain.

2. Watch this video in which Van Jacobson discusses the congestion collapse of the 1980s and how he figured out how to fix it https://www.youtube.com/watch?v=QP4A6L7CEqA

Similar article: https://www.wired.com/2012/05/van-jacobson/

Links to an external site.

What problem is he describing and what techniques did he employ to solve the problem? Discuss and explain.

- The problem he is describing is that in 1980, the internet experienced a congestion collapse which meant that as more packets were injected into the internet the throughput dramatically decreased. Some techniques Van Jacobson employed to solve this problem were: Slow Start, Congestion Avoidance, Additive Increase/Multiplicative Decrease, and RTT estimation. Jacobson's work helped make the internet scalable and reliable by managing congestion efficiently which prevented future collapse.

3. TCP Questions:

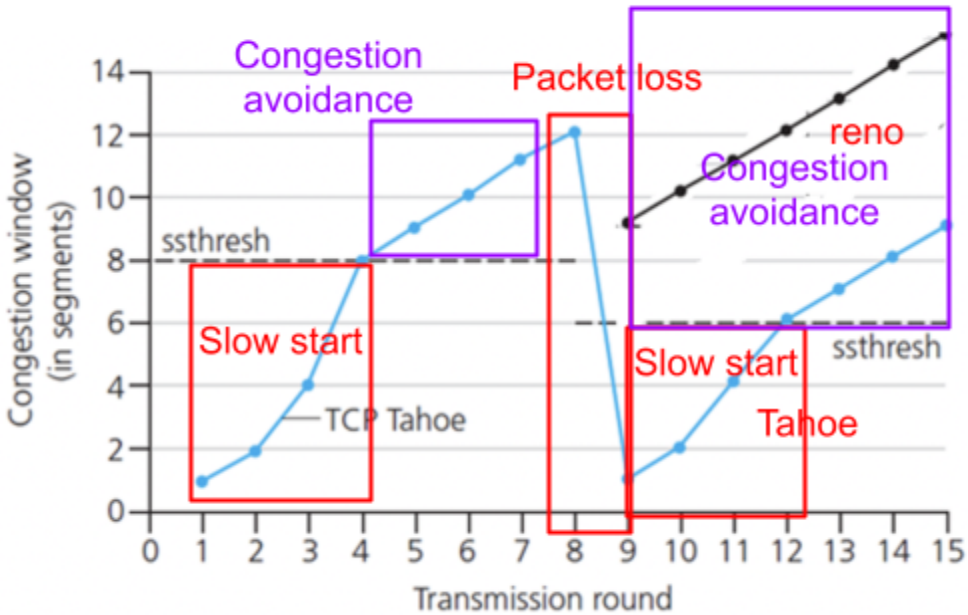
1. Why are the congestion control algorithms in TCP important?
TCP congestion control algorithms are important because they ensure stability and efficiency of data transmission across the network. They dynamically adjust the transmission rate based on network conditions, optimizing throughput while minimizing packet loss and delays. Without these algorithms, excessive traffic could overwhelm the network, leading to inefficiency and degraded performance.

2. What causes the "sawtooth" behavior that we notice in AIMD?
The "sawtooth" behavior arises from TCP's cycle of increasing and reducing its congestion window (aka. `cwnd`). During normal operation, the window grows linearly until packet loss signals congestion. When this occurs, TCP halves the window size, creating a sharp drop. This cycle of growth and reduction creates the "sawtooth" behavior.

3. For the graph below, identify (mark up) the graph to show the following:

- packet loss
- period(s) of slow start

- period(s) of congestion avoidance
- After the packet loss:
    - TCP Tahoe
    - TCP Reno



4. In the graph above, suppose MSS = 1Kbyte

- How much data has been transmitted at the end of the 7th transmission round?
  (1+2+4+8+9+10+11) * 1 Kbyte = 45 Kbyte
- How much data is transferred in the transmission round following the packet loss with
    - --TCP Reno?
        - cwnd = 12 / 2 + 3 = 9
        - Data transferred = 9 * 1 KBytes = 9 Kbytes
    - --TCP Tahoe?
        - cwnd = 1
        - Data transferred = 1 * 1 KBytes = 1 Kbytes

  (show your work)

- Mark the point where there is packet loss. Look over the window adjustment rules and explain the difference in the adjustments made by Tahoe compared to Reno. Use the lecture slide as a guide and mark up the graph showing a) packet loss b) Tahoe adjustment and c) Reno adjustment.

In both Reno and Tahoe, ssthresh is set to half the congestion window size when packet loss occurs. In Reno, the congestion window becomes half of the previous congestion window size plus three while in Tahoe, the congestion window size is reset to 1 MSS. Reno recovers faster than Tahoe since it skips the slow start phase and immediately resumes from the reduced ssthresh in congestion avoidance.

5. What is the benefit of TCP's Fast Restransmit algorithm?

- TCP's Fast retransmit algorithm improves data transmission efficiently by detecting and restransmitting the lost packets upon receiving multiple duplicate ACKs without waiting for a timeout. This helps reduce recovery time, and helps maintain throughput without unnecessary congestion allowing for smoother and faster recovery from packet loss.

Does the Fast Restransmit take any steps to mitigate congestion in the network?

- Fast Retransmit does take steps to mitigate network congestion by reducing the congestion window (cwnd) once it detects packet loss. This limits data flow to prevent further congestion and then it enters fast recovery to mainted transmission to balance recovery and network stability.